

Chronic kidney disease (CKD)

NICE NG203, 2021



Red Whale

GEMS

Guidelines & Evidence Made Simple

Diagnosis

Diagnose CKD if any of:

- eGFR <60 on 2 occasions at least 3 months apart
- ACR ≥3 (repeat on early morning sample if 3–70)
- Structural or genetic kidney diseases

If eGFR low, redo within 2 weeks to exclude AKI!

Consider targeted screening for CKD if:

- Previous AKI
- Incidental dip finds blood or protein
- Diabetes
- Hypertension or CVD
- Taking certain drugs, e.g. long-term NSAIDs
- Prostatic hypertrophy or renal stones

Investigations

- U&Es
- FBC (esp. if eGFR ≤30)
- ACR
- Urine dip (for blood)
- LFT and bone profile
- BP

Ultrasound if:

- eGFR <30
- Accelerated CKD (see definition below)
- Haematuria (visible or persistent non-visible)
- Symptoms suggesting obstruction
- Age >20y and family Hx polycystic kidneys

Don't dip urine for protein



DO dip urine for BLOOD
DO send urine for ACR

Calculate risk of needing renal replacement

NEW!

Calculate 5-year risk of needing renal replacement using 4-variable kidney failure risk equation. Refer if 5-year risk is >5%

How? Download from NICE website (under 'Implementation support'):
<https://www.nice.org.uk/guidance/ng203/resources>

Classification

G1	G2	G3a	G3b	G4	G5
eGFR >90	eGFR 60–89	eGFR 45–59	eGFR 30–44	eGFR 15–29	eGFR <15



Management: think ALPHA

A

Atorvastatin 20mg for ALL (primary AND secondary prevention)

- 1° prevention: aim for >40% reduction in non-HDL cholesterol
 - 2° prevention: aim for LDL ≤2mmol/l or non-HDL ≤2.6mmol/l
- If targets not met, increase dose if eGFR ≥30 (if eGFR <30, do so on advice of renal specialist) (NICE 2023, NG238)**

L

Lifestyle, especially (but not only!):

- Optimisation of BMI
- Smoking cessation

P

Proteinuria

- ACEi/ARB (and up-titrate) if ACR ≥70, or ACR ≥3 if diabetes
- GLIFLOZIN (once on maximum tolerated dose of ACEi/ARB)
 - OFFER if T2DM and ACR >30
 - Consider if T2DM and ACR 3–30*

Above is from NICE GUIDELINE. Technology appraisals outline other situations where dapag and empag are 'an option' (see main article)

H

Hypertension: manage as per NICE hypertension guidance

Targets are based on ACR, regardless of age:

ACR <70:
target clinic BP is
<140/90

ACR ≥70:
target clinic BP is
<130/80

A

Anaemia: investigate and manage, but note:

- eGFR >60: CKD unlikely to be the cause
- eGFR 30–60: CKD may be the cause
- eGFR <30: CKD often the cause

Monitoring frequency

ANNUAL	eGFR ≥45 and ACR ≤30
6 MONTHLY	eGFR 30–59 and ACR >30 eGFR 15–45 and ACR ≤30
4 MONTHLY	eGFR 15–29 and ACR >30
3 MONTHLY	eGFR <15 (any ACR)

Referral

- 5-year risk >5% (on 4-variable calculator)
- Accelerated CKD (see below)
- ACR ≥70 (unless due to diabetes and treated)
- ACR >30 plus haematuria
- CKD + renal outflow obstruction (urology)
- CKD + hypertension resistant to 4 agents
- Structural/genetic causes of CKD



Accelerated CKD
Over 12 months:

Absolute sustained fall in eGFR of 15ml/min
OR

Relative eGFR fall ≥25% which also results in a worse CKD staging

We make every effort to ensure the information in these pages is accurate and correct at the date of publication, but it is of necessity of a brief and general nature. The information presented herein should not replace your own good clinical judgement, or be regarded as a substitute for taking professional advice in appropriate circumstances. In particular, we suggest you carefully consider the specific facts, circumstances and medical history of any patient, and recommendations of the relevant regulatory authorities. We also suggest that you check drug doses, potential side-effects and interactions with the British National Formulary. Save insofar as any such liability cannot be excluded at law, we do not accept any liability for loss of any type caused by reliance on the information in these pages. January 2024. For full references see the relevant Red Whale articles.